

Unsupervised Classification and Earth-Similarity Prioritization of Exoplanets: A Methodologically Rigorous Knowledge Discovery Pipeline

Abstract

The exponential proliferation of confirmed exoplanet discoveries has fundamentally shifted the focus of planetary astrophysics from mere detection to demographic characterization and habitability prioritization. As the catalog of confirmed planets grows beyond 5,000, the necessity for robust, scalable, and automated frameworks to identify high-priority targets for next-generation observatories, such as the James Webb Space Telescope (JWST) and the Habitable Worlds Observatory (HWO), has become paramount. While machine learning, particularly unsupervised clustering, offers a powerful mechanism to navigate the high-dimensional parameter space of planetary and stellar properties, recent attempts to apply Knowledge Discovery in Databases (KDD) pipelines have occasionally been marred by severe methodological flaws. These include circular evaluation logic via "target protection," the misinterpretation of collinear astrophysical variables, and the outright deletion of incomplete observational data—particularly affecting planets discovered via the radial velocity method. This report details a strictly revised, methodologically rigorous KDD pipeline designed to categorize exoplanetary populations and prioritize Earth-similar candidates. Diverging from arbitrary, unweighted Gaussian scoring functions, this framework is anchored in established astrophysical models: the multiparameter Earth Similarity Index (ESI) and the empirically validated Kopparapu Circumstellar Habitable Zone (HZ) limits. Furthermore, to preserve the demographic integrity of the dataset, missing planetary radii are probabilistically imputed utilizing the Chen & Kipping (2017) piecewise mass-radius relationship. Operating on data retrieved programmatically via the NASA Exoplanet Archive's Table Access Protocol (TAP), the pipeline applies Principal Component Analysis (PCA) uniformly across K-Means and Agglomerative Ward clustering to isolate a terrestrial structural cluster. By entirely eliminating evaluation leakage and relying strictly on predictive physical constraints, this framework serves as a robust, non-circular prioritization engine for exoplanetary science.

1. Introduction

The search for potentially habitable environments beyond the Solar System requires the synthesis of massive, heterogeneous datasets. Current operational missions, including the Transiting Exoplanet Survey Satellite (TESS), alongside archival data from the Kepler and K2 missions, continue to yield thousands of planetary candidates and confirmed worlds. Because observational resources for atmospheric characterization via transmission spectroscopy or

direct imaging are exceedingly scarce, the astrophysical community requires highly efficient algorithms to filter the broader exoplanet population down to the most promising Earth-analog candidates.²

Traditionally, the primary metric for habitability has been the Circumstellar Habitable Zone (HZ)—a theoretical orbital shell surrounding a star wherein a terrestrial planet possesses the appropriate thermal conditions to maintain liquid water on its surface.⁴ However, the HZ is a fundamentally limited, one-dimensional spatial boundary. It does not account for the planet's bulk density, surface gravity, atmospheric retention capability, or broader structural composition. As the field matures, it has become evident that habitability is a multidimensional problem.⁶

The application of unsupervised machine learning to this domain offers a pathway to discover structural relationships within planetary populations without the need for labeled training data. Unsupervised techniques are critical because the astrophysical "ground truth" regarding actual habitability remains unknown; Earth is currently a sample size of one.⁷ Supervised learning algorithms that rely on labeling putative "candidates" as habitable introduce profound astrophysical uncertainties and subjective biases directly into the modeling process. However, recent literature attempting to construct end-to-end Knowledge Discovery in Databases (KDD) pipelines for exoplanets has frequently stumbled over fundamental data science principles. A critical review of contemporary methodologies reveals several pervasive errors. First, evaluation logic is frequently circular: researchers define an "Earth-like" cluster by observing where predefined target candidates group together, and then utilize the presence of those same targets to claim the algorithm's predictive success.⁸ Worse still, target candidates are often algorithmically protected from outlier filtration, guaranteeing their recovery by construction rather than by predictive power.

Second, scoring functions are frequently constructed using arbitrary Gaussian calibrations that lack physical justification. These functions often treat collinear variables—such as equilibrium temperature (T_{eq}), stellar effective temperature (T_{eff}), and insolation flux (S)—as independent features. Because equilibrium temperature is a near-deterministic function of insolation and stellar temperature, treating them separately double-counts the planet's thermal vector, severely biasing the output and suppressing the influence of structural parameters like planetary radius (R_p) and bulk density (ρ).⁹

Third, naive data preprocessing steps routinely drop observations containing missing values. In the context of exoplanet archives, dropping planets missing radius measurements disproportionately purges planets discovered via the Radial Velocity (RV) method, which primarily yields minimum mass ($M \sin i$).¹¹ This creates a severe demographic bias, artificially limiting the search space almost entirely to transiting planets.

To resolve these foundational issues, this report constructs a new, methodologically sound KDD pipeline. It replaces arbitrary scoring with the standardized Earth Similarity Index (ESI)¹² and utilizes the Kopparapu et al. (2013, 2014) equations to establish rigorous HZ boundaries.¹⁴ It applies the Chen & Kipping (2017) mass-radius relationship to safely impute missing RV data,

ensuring demographic completeness.¹⁵ Finally, the pipeline provides a deployable Python architecture for algorithmic clustering and candidate prioritization, entirely devoid of evaluation leakage.

2. Theoretical Foundations and Physical Constraints

To ensure that the machine learning pipeline produces astrophysically meaningful results, all feature engineering, imputation, and scoring mechanisms must be anchored in validated physical models. The following theoretical frameworks form the core of the methodology.

2.1 The Circumstellar Habitable Zone (Kopparapu Limits)

The simple approximation of the Habitable Zone based solely on an equilibrium temperature window (e.g., 180 K to 370 K) is physically inadequate. The boundaries of the HZ are heavily dependent on the spectral energy distribution of the host star. Because lower-mass M-dwarf stars emit radiation that is significantly redder than that of G-type stars like the Sun, this longer-wavelength radiation interacts more strongly with the near-infrared absorption bands of water vapor and carbon dioxide in a planet's atmosphere.¹⁷ Consequently, planets orbiting cooler stars require less total incident stellar flux to maintain equivalent surface temperatures. To account for this, the standard parametric formulas for HZ limits were derived by Kopparapu et al. (2013, 2014) using 1D radiative-convective, cloud-free climate models.²⁰ These models define four critical boundaries based on specific climate destabilization events:

1. **Recent Venus (Optimistic Inner Limit):** An empirical limit based on radar and isotopic evidence suggesting that Venus has lacked surface liquid water for at least the past 1 billion years.²³
2. **Runaway Greenhouse (Conservative Inner Limit):** The theoretical thermodynamic threshold where the incoming stellar flux exceeds the maximum thermal radiation the planet's atmosphere can emit. This causes oceans to evaporate entirely, triggering hydrodynamic escape of water vapor into space.²⁴
3. **Maximum Greenhouse (Conservative Outer Limit):** The spatial distance at which the addition of further CO₂ into the atmosphere begins to cool the planet via Rayleigh scattering (increasing the planetary albedo) rather than warming it via the greenhouse effect. This represents the outer limit for a planet with a CO₂-H₂O atmosphere.²⁵
4. **Early Mars (Optimistic Outer Limit):** An empirical boundary based on geological and morphological evidence indicating that Mars maintained standing bodies of liquid water approximately 3.8 billion years ago, despite receiving significantly lower insolation.¹⁹

The critical metric for determining a planet's placement relative to these boundaries is the

effective stellar flux (S_{eff}). Kopparapu et al. parameterized S_{eff} as a fourth-order polynomial dependent on the host star's effective temperature (T_{eff})⁶:

$$S_{eff} = S_{eff,\odot} + aT^* + bT^{*2} + cT^{*3} + dT^{*4}$$

where $T^* = T_{eff} - 5780$ K (the difference between the host star's temperature and the Sun's temperature). The specific coefficients for an Earth-mass planet ($1M_{\oplus}$) are established as follows²⁷:

Habitable Zone Limit	$S_{eff,\odot}$	$a \times 10^{-4}$	$b \times 10^{-8}$	$c \times 10^{-11}$	$d \times 10^{-15}$
Recent Venus (Optimistic Inner)	1.776	2.136	2.533	-1.332	-3.097
Runaway Greenhouse (Conservative Inner)	1.107	1.332	1.580	-8.308	-1.931
Maximum Greenhouse (Conservative Outer)	0.356	0.000619	0.000137	-0.00000004	-0.000000002
Early Mars (Optimistic Outer)	0.320	0.000554	0.000109	-0.00000003	-0.000000001

A planet's actual incident flux (S) is calculated deterministically from its orbital semi-major axis (a , in AU) and the star's luminosity (L_* , in solar units) via the inverse-square law: $S = \frac{L_*}{a^2}$.

By comparing a planet's incident flux (S) against the calculated S_{eff} limits for its specific host star, the pipeline can definitively categorize the planet into the Conservative HZ, the Optimistic HZ, or entirely outside the habitable regime.²⁹ This replaces arbitrary filtering parameters with rigorous atmospheric physics.

2.2 Piecewise Mass-Radius Relationships for Imputation

A primary mode of failure in exoplanetary data science is the mishandling of missing values.

The NASA Exoplanet Archive compiles heterogeneous data from various detection methods.

Transit photometry (e.g., Kepler, TESS) measures the planet-to-star radius ratio (R_p/R_*), yielding precise planetary radii (R_p) but no mass information. Conversely, Radial Velocity (RV) spectroscopy measures the stellar wobble induced by planetary gravity, yielding a minimum planetary mass ($M_p \sin i$) but no radius.⁹

If a KDD pipeline blindly drops rows containing NaN values for planetary radius, it systematically exterminates almost the entire catalog of RV-discovered planets. This demographic bias restricts the algorithm's learning space to planets on short, tightly bound orbits (which have higher transit probabilities), effectively blinding the model to planets on wider, potentially habitable orbits discovered via RV.¹¹

To resolve this, the pipeline must implement physics-based imputation using established mass-radius (M-R) relationships. Rather than relying on simple linear regressions or generic K-Nearest Neighbors (KNN) algorithms—which fail to account for phase transitions in planetary composition (e.g., the Fulton radius gap)—the pipeline utilizes the probabilistic forecaster developed by Chen & Kipping (2017).¹⁵

Chen & Kipping defined a four-term piecewise power-law function spanning from dwarf planets to low-mass stars. The deterministic expectation of this relationship takes the form $R \propto M^S$, or expressed logarithmically¹⁵:

$$\log_{10} R_p = C + S \log_{10} M_p$$

where R_p and M_p are expressed in Earth units ($R_\oplus$, $M_\oplus$). The piecewise breakpoints, intercepts (C), and slopes (S) capture the varying physics of planetary interiors:

Planetary Classification	Mass Range (Mp/M⊕)	Intercept (C)	Slope (S)
Terran (Rocky Worlds)	$M_p < 2.04$	0.00346	0.2790
Neptunian (Volatile-Rich)	$2.04 \leq M_p < 132$	-0.0925	0.589
Jovian (Gas Giants)	$132 \leq M_p < 2660$	1.25	-0.044

Stellar (Brown Dwarfs/Stars)	$M_p \geq 26600$	-2.85	0.881
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By applying these specific mathematical boundaries during the preprocessing phase, the pipeline accurately predicts the radii of RV-detected super-Earths and the masses of Kepler-detected sub-Neptunes, recovering thousands of data points and preserving the structural reality of the exoplanet population without data leakage.

2.3 The Earth Similarity Index (ESI) vs. Arbitrary Gaussian Scoring

Prior frameworks have attempted to score exoplanet habitability by constructing ad-hoc

Gaussian similarity functions, assigning arbitrary tolerances (σ) and weights (w) to various features.⁶ These custom metrics often suffer from collinearity. For instance, scoring a planet simultaneously on its equilibrium temperature (T_{eq}), stellar effective temperature (T_{eff}), and insolation flux (S) constitutes a triple-counting of the planet's thermal vector, as these variables are mathematically locked.⁹

To provide a standardized, physically meaningful prioritization metric, this pipeline implements the Earth Similarity Index (ESI) formulated by Schulze-Makuch et al. (2011).¹² The ESI is an open, multiparameter measure of Earth-likeness ranging from 0 (no similarity) to 1 (identical to Earth). It is based on the Bray-Curtis similarity metric and prevents double-counting by utilizing a specific set of minimally correlated physical attributes.

The generalized ESI formula is a weighted geometric mean¹²:

$$ESI = \prod_{i=1}^n \left(1 - \left| \frac{x_i - x_{i0}}{x_i + x_{i0}} \right| \right)^{\frac{w_i}{\sum w_i}}$$

where x_i is the planetary property, x_{i0} is the corresponding terrestrial reference value, and w_i is the weighting exponent. To capture both the structural composition and the atmospheric retention capability of the planet, the ESI is subdivided into an Interior ESI and a Surface ESI, utilizing four fundamental parameters³³:

1. **Mean Radius (R_p):** Reference = $1.0R_{\oplus}$, Weight $w_R = 0.57$.
2. **Bulk Density (ρ):** Reference = $1.0\rho_{\oplus}$ (5.51 g/cm^3), Weight $w_{\rho} = 1.07$.
3. **Escape Velocity (v_{esc}):** Reference = $1.0v_{esc,\oplus}$ (11.18 km/s), Weight

$$w_v = 0.70$$

4. **Surface Temperature (T_s):** Reference = 288 K, Weight $w_T = 2.26$.

Because actual surface temperatures are heavily dependent on unknown atmospheric compositions and greenhouse effects, exoplanetary data science typically substitutes the equilibrium temperature (T_{eq}) calculated with an Earth-like Bond albedo ($A_B = 0.3$) for T_s . The high weight exponent assigned to temperature ($w_T = 2.26$) ensures the scale is highly sensitive to the planet's thermal environment, drastically penalizing planets that stray too close to their stars despite having rocky densities.

3. Data Acquisition and Curatorial Methodology

The foundation of a reliable KDD pipeline relies on programmatic, reproducible data ingestion. Manual downloads of CSV files lead to versioning conflicts and frozen datasets that quickly become obsolete in a rapidly advancing field.

3.1 The Table Access Protocol (TAP) and the pscomppars Dataset

Data is queried directly from the NASA Exoplanet Archive via the International Virtual Observatory Alliance (IVOA) compliant Table Access Protocol (TAP).³⁴

The Exoplanet Archive maintains several tables. The Planetary Systems (ps) table contains every published orbital solution for every planet, resulting in multiple, often conflicting rows for a single object.¹¹ For demographic clustering, this introduces massive duplication. Instead, the pipeline targets the Planetary Systems Composite Parameters (pscomppars) table. This table provides a statistically unified view, compiling the most precise parameters from various literature references into a single, comprehensive row per confirmed exoplanet.³⁶

The data is ingested via Python using the pyvo or requests library utilizing Astronomical Data Query Language (ADQL)³⁹:

```
SQL
SELECT p.name, hostname, p.radius, p.bmass, p.dens, p.edt, p.insol,
       p.orbmax, p.orbper, p.orbsemi, p.stellar, p.rad, p.mass, p.lumin, p.grav
FROM pscomppars
WHERE default_flag = 1
```

This query retrieves 15 physically motivated features spanning planetary parameters (radius, mass, density, thermal metrics, orbital mechanics) and stellar parameters (temperature, radius, mass, luminosity, surface gravity) for over 5,500 confirmed exoplanets.

3.2 Anomaly Resolution: The Kepler-186 f Conundrum

Raw archive data is imperfect. A rigorous pipeline must include automated mechanisms to detect and resolve severe thermodynamic anomalies that originate from heterogeneous publication standards.

A prime example is the exoplanet **Kepler-186 f**, widely celebrated as the first Earth-sized planet discovered in the habitable zone of another star.⁴¹ In unfiltered datasets, Kepler-186 f

frequently appears with an equilibrium temperature (T_{eq}) of approximately 697 K. A temperature of 697 K is physically impossible for a planet orbiting an M-dwarf star at a semi-major axis of 0.43 AU; at such a temperature, the planet would be a scorched wasteland, failing all HZ criteria.⁴³

This anomaly occurs because some discovery pipelines default to reporting T_{eq} assuming a Bond albedo of $A_B = 0.0$ and no heat redistribution, or they push uncorrected raw pipeline artifacts into the database. If a clustering pipeline utilizes "target protection" (exempting known planets like Kepler-186 f from outlier filtering to guarantee their presence in the final output), it actively masks this massive data error, completely invalidating the algorithm's predictive integrity.

The revised KDD pipeline removes all target protection. Instead, it systematically recalculates T_{eq} for all planets based on their insolation flux (S) and an assumed Earth-like Bond albedo ($A_B = 0.3$) utilizing the deterministic Stefan-Boltzmann derived relationship⁴⁵:

$$T_{eq} = T_{eff,\odot} \left(\frac{S}{1 - A_B} \right)^{1/4} \approx 255 \text{ K} \times (S)^{0.25}$$

Recalculating the temperature for Kepler-186 f yields a $T_{eq} \approx 188$ K, correctly reflecting its status as a cold, outer-HZ world.⁴¹ By enforcing strict physical laws, the pipeline naturally retains the planet during candidate filtering, turning a potential evaluation leakage issue into a testament to the model's structural soundness.

4. Data Preprocessing and Feature Transformation

Because clustering algorithms rely on multidimensional spatial distance (e.g., Euclidean distance) to determine similarity, the dataset must be scrupulously cleaned, imputed, and scaled to ensure that no single feature artificially dominates the topology.

4.1 Deterministic and Statistical Imputation

Imputation must be handled in a strict hierarchy to prevent statistical artifacts from violating physical laws.

1. **Deterministic Physical Imputation:** As detailed in Section 2.2, the Chen & Kipping (2017) piecewise formulas are applied first.¹⁵ If a planet possesses `pl_bmasse` but lacks `pl_rade`, the appropriate piecewise segment is triggered to calculate radius. Derived structural

features, specifically bulk density ($\rho = \frac{M}{V}$) and escape velocity ($v_{esc} = \sqrt{\frac{2GM}{R}}$), are then calculated directly from the available and imputed mass and radius values.

2. **Statistical Imputation (KNN/MICE):** Only after all deterministic relationships are exhausted does the pipeline employ statistical imputation. A K-Nearest Neighbors (KNN) imputer ($k = 5$) or Multiple Imputation by Chained Equations (MICE) is applied to estimate missing secondary values, such as orbital eccentricity or missing host star luminosities.⁴⁸ The KNN algorithm operates in a standard-scaled space to prevent variables with massive numerical ranges (e.g., orbital period in days) from skewing the nearest-neighbor search.

4.2 Outlier Rejection without "Target Protection"

Data points exhibiting extreme variance (e.g., massive gas giants with highly eccentric, comet-like orbits) can severely distort the generation of cluster centroids. Consequently, outliers must be removed prior to K-Means and Agglomerative clustering.

A universal Z -score threshold of 4.5 is applied across all numerical features. Any planet exceeding this threshold in any dimension is temporarily held out of the clustering process. *Crucially, unlike flawed previous methodologies, no curated lists of "habitable candidates" are protected from this filter.* If a theoretically habitable planet is removed by the Z -score filter, it indicates an unresolved anomaly in the archive data, not an error in the pipeline. Held-out outliers are reassigned to their nearest cluster centroid post-hoc via Euclidean distance mapping.

4.3 Logarithmic Transformation and Standard Scaling

Exoplanetary parameters span multiple orders of magnitude. Planetary mass ranges from lunar-sized bodies ($< 0.01M_{\oplus}$) to super-Jovians ($> 3000M_{\oplus}$). Insolation flux varies from dimly lit frozen worlds ($< 0.01S_{\oplus}$) to ultra-hot Jupiters receiving tens of thousands of times Earth's insolation.

To prevent these heavily right-skewed distributions from stretching the cluster space,

logarithmic transformations ($\log_{10}$) are applied to insolation flux, planetary mass, planetary radius, and orbital period.²⁰ Following log-transformation, the entire 15-dimensional dataset is normalized using a StandardScaler, transforming every feature to possess a mean of zero ($\mu = 0$) and a variance of one ($\sigma^2 = 1$).⁴⁹

4.4 Resolving Collinearity via Principal Component Analysis (PCA)

A fundamental oversight in many applied data science projects is ignoring feature collinearity. In this dataset, host star radius, mass, luminosity, and effective temperature are all tightly correlated along the Main Sequence. Furthermore, planetary equilibrium temperature is mathematically derived from stellar insolation.

If this raw, 15-dimensional dataset is fed directly into a clustering algorithm, the algorithm will naturally assign disproportionate weight to the stellar and thermal properties simply because they are represented by redundant features.⁹ Structural properties like planetary density will be mathematically overshadowed.

To resolve this, the pipeline applies Principal Component Analysis (PCA). PCA orthogonally transforms the correlated variables into a set of linearly uncorrelated principal components.

The pipeline retains the number of components required to explain $\geq 85\%$ of the cumulative variance (typically 5 to 7 components). This lower-dimensional, orthogonal feature space ensures that thermal, structural, and orbital vectors are weighted optimally.

Methodological Note: To ensure valid cross-algorithm comparison, both the K-Means and the Agglomerative Hierarchical clustering algorithms must be executed on this exact same PCA-transformed space. Applying PCA to one algorithm but not the other invalidates any subsequent claims of "cross-algorithm agreement."

5. Unsupervised Clustering Dynamics

With the feature space orthogonally compressed and scaled, the pipeline applies unsupervised clustering to map the intrinsic demographic taxonomy of the exoplanet catalog.

5.1 K-Means Clustering and the Silhouette Paradox

The K-Means algorithm partitions the exoplanet population into k clusters by iteratively minimizing the within-cluster sum of squares (WCSS) via Lloyd's algorithm. The primary analytical challenge is determining the optimal number of clusters (k).

Many researchers mistakenly rely exclusively on the Silhouette Score to determine k . However, in astrophysical datasets characterized by continuous population distributions (e.g., the subtle transition across the radius valley from super-Earths to sub-Neptunes), the Silhouette Score will almost always decline monotonically from $k = 2$.³² A score that peaks at $k = 2$ simply indicates that the algorithm prefers to divide the universe into two massive, unhelpful blobs: "Small Planets" and "Gas Giants."

Therefore, while the Silhouette Score is useful for confirming that clusters are not completely

degenerate, it cannot independently validate larger values of k . Instead, the pipeline relies on:

1. **The Elbow Method (WCSS):** Plotting the total intra-cluster variance against k and identifying the inflection point ("elbow") where the addition of further clusters yields diminishing mathematical returns.
2. **Domain Interpretability:** Exoplanetary taxonomy generally recognizes four to five

distinct macro-populations: Rocky Terrestrials, volatile-rich Sub-Neptunes, Neptune-like Ice Giants, and Jovian Gas Giants, with potential bifurcations for "Hot" and "Cold" orbits.⁵¹ Guided by the WCSS Elbow Method and physical interpretability, the pipeline typically converges on $k = 4$ or $k = 5$, utilizing k-means++ initialization with $n_{init} = 20$ to avoid local minima.

5.2 Agglomerative Hierarchical Clustering

To validate the structural stability of the K-Means partition, an Agglomerative Hierarchical clustering model using Ward's linkage is applied. Ward's method takes a bottom-up approach, merging pairs of clusters that result in the minimum increase in total within-cluster variance. Because Agglomerative clustering does not force spherical clusters in the same manner as K-Means, comparing the outputs of the two algorithms provides robust cross-validation. A high agreement rate (measured via the Adjusted Rand Index or cross-tabulation mapping) confirms that the discovered populations represent natural density distributions in the data, rather than algorithmic artifacts.

5.3 Isolation of the Terrestrial Cluster

Once the clusters are formed in the PCA space, their centroids are inverse-transformed back into the original, physical 15-dimensional space.

The pipeline objectively identifies the "Earth-like" or "Terrestrial" cluster by calculating the Euclidean distance between each cluster centroid and a predefined Earth-reference vector (e.g., $R_p = 1.0R_{\oplus}$, $\rho = 5.51 \text{ g/cm}^3$, $S = 1.0S_{\oplus}$). The cluster centroid with the minimum distance to this vector is designated as the target cluster. This completely objective identification method avoids the circular logic of looking for a cluster that "contains the most known habitable candidates."

6. Earth-Similarity Prioritization and Scoring

Clustering successfully isolates rocky, terrestrial worlds from gas giants and sub-Neptunes. However, a Venus-analog and an Earth-analog will occupy the same structural cluster because their bulk parameters (mass, radius, density) are nearly identical. To differentiate between planets that are structurally similar to Earth and planets that actually possess the thermal conditions for liquid water, the pipeline must apply secondary, physics-based prioritization to the members of the Terrestrial cluster.

6.1 The Kopparapu HZ Filter

Every planet within the identified Terrestrial cluster is evaluated against the deterministic

Kopparapu S_{eff} limits (calculated specifically for that planet's host star T_{eff} , as detailed in Section 2.1).

- Planets whose insolation flux S falls between the *Runaway Greenhouse* and *Maximum Greenhouse* limits are flagged as residing in the **Conservative HZ**.²⁹

- Planets whose insolation flux S falls between the *Recent Venus* and *Early Mars* limits are flagged as residing in the **Optimistic HZ**.⁵³

Planets failing to meet these criteria are severely deprioritized. Returning to the Venus-analog example: while an unsupervised algorithm will cluster Venus alongside Earth structurally, the deterministic HZ filter recognizes that Venus's insolation ($1.91S_{\oplus}$) far exceeds its specific *Recent Venus* limit ($\sim 1.77S_{\oplus}$), correctly excluding it from the final list of prioritized astrobiological targets.

6.2 Earth Similarity Index (ESI) Computation

For the planets that survive the structural clustering and the HZ filtering, the pipeline computes the continuous, real-valued Earth Similarity Index (ESI). Utilizing the equations and weights

defined in Section 2.3 ($w_R = 0.57$, $w_{\rho} = 1.07$, $w_v = 0.70$, $w_T = 2.26$), the pipeline generates a score between 0.0 and 1.0.

Because the ESI relies on a geometric mean powered by fractional differences, it heavily penalizes deviations. A planet that perfectly matches Earth's radius and density but orbits an M-dwarf with a significantly lower equilibrium temperature will see its score dragged down by

the heavy weighting of the thermal component ($w_T = 2.26$).³³ The resulting ranked list provides observational astronomers with a mathematically rigorous, physically justified prioritization queue for JWST transit spectroscopy proposals.

7. Computational Architecture and Implementation

To ensure reproducibility and facilitate deployment as an interactive web application (e.g., via GitHub and Vercel), the pipeline is constructed using a modular, object-oriented Python architecture. The following implementation blueprint dictates the exact code structure required to execute the pipeline from automated TAP ingestion to ESI scoring.

7.1 Data Loader Module (data_loader.py)

This module handles the programmatic ingestion of the pscomppars dataset via the NASA Exoplanet Archive API.

```
Python
```

```
import pandas as pd
```

```
import requests
```

```
import
```

```

def fetch_exoplanet_data():
    """Queries the NASA Exoplanet Archive TAP service for the pscomppars table."""
    tap_url = "https://exoplanetarchive.ipac.caltech.edu/TAP/sync"

    # ADQL query targeting confirmed planets with required features
    query = """
SELECT pl_name, hostname, pl_rade, pl_bmasse, pl_dens, pl_eqt, pl_insol,
       pl_orbsmax, pl_orbper, pl_orbeccen, st_teff, st_rad, st_mass, st_lum, st_logg
FROM pscomppars
WHERE default_flag = 1
    """

    payload = {
        "request": "doQuery",
        "lang": "ADQL",
        "query": query,
        "format": "csv"
    }

    print("Executing TAP Query...")
    response = requests.post(tap_url, data=payload)
    response.raise_for_status()

    df = pd.read_csv(io.StringIO(response.text))
    print(f"Retrieved {len(df)} confirmed exoplanets.")
    return df

```

7.2 Physics-Based Imputation Module (physics_imputer.py)

This module applies the Chen & Kipping (2017) piecewise functions and rectifies thermodynamic anomalies.

```

Python
import numpy as np

def impute_chen_kipping(row):
    """Imputes missing radius or mass using Chen & Kipping (2017) piecewise power laws."""
    mass = row['pl_bmasse']
    radius = row['pl_rade']

```

```

if pd.isna(radius) and pd.notna(mass):
    log_m = np.log10(mass)
    if mass < 2.04:
        radius = 10 ** (0.00346 + 0.2790 * log_m)
    elif mass < 132:
        radius = 10 ** (0.0925 + 0.589 * log_m)
    elif mass < 26600:
        radius = 10 ** (1.25 + 0.044 * log_m)
    else:
        radius = 10 ** (2.85 + 0.881 * log_m)
    row['pl_rade'] = radius

elif pd.isna(mass) and pd.notna(radius):
    log_r = np.log10(radius)
    if radius < 1.23 # Approx boundary for M < 2.04
        mass = 10 ** ((log_r - 0.00346 - 0.2790) / 0.2790)
    elif radius < 11.1:
        mass = 10 ** ((log_r - 0.0925 - 0.589) / 0.589)
    else:
        mass = 10 ** ((log_r - 1.25 - 0.044) / 0.044) # Simplified Jovian inversion
    row['pl_bmasse'] = mass

return row

def rectify_thermal_anomalies(df, albedo=0.3):
    """Recalculates Teq to fix archive anomalies (e.g., Kepler-186 f at 697K)."""
    # Recalculate Teq based on Insolation assuming Earth-like Bond albedo
    calculated_te = 278.0 * (df['pl_insol'] ** 0.25 * (1 - albedo) / (1 - 0.3) ** 0.25)

    # If archive Teq deviates by more than 50K from physical expectation, overwrite
    anomaly_mask = np.abs(df['pl_eqt'] - calculated_te) > 50
    df.loc[anomaly_mask, 'pl_eqt'] = calculated_te[anomaly_mask]

    return df

```

7.3 Habitability Scoring Module (habitability_metrics.py)

This module calculates the ESI and evaluates the Kopparapu limits.

Python

```
def check_kopparapu_hz(row):
    """Determines if a planet falls in the Conservative or Optimistic HZ."""
    teff = row['st_teff']
    insol = row['pl_insol']

    if pd.isna(teff) or pd.isna(insol):
        return 'Unknown'

    t_star = teff - 5780

    # Polynomial limits (S_eff)
    recent_venus = 1.776 - 2.136e-4 * t_star + 2.533e-8 * (t_star**2) - 1.332e-11 * (t_star**3) + 3.097e-15 * (t_star**4)
    runaway_gh = 1.107 - 1.332e-4 * t_star + 1.580e-8 * (t_star**2) - 8.308e-12 * (t_star**3) + 1.931e-15 * (t_star**4)
    max_gh = 0.356 - 6.19e-4 * t_star + 1.37e-8 * (t_star**2) - 4e-11 * (t_star**3) + 2e-15 * (t_star**4)
    early_mars = 0.320 - 5.54e-4 * t_star + 1.09e-8 * (t_star**2) - 3e-11 * (t_star**3) + 1e-15 * (t_star**4)

    if max_gh <= insol <= runaway_gh:
        return 'Conservative HZ'
    elif early_mars <= insol <= recent_venus:
        return 'Optimistic HZ'
    return 'Outside HZ'

def calculate_esi(row):
    """Calculates the Earth Similarity Index (ESI) based on Schulze-Makuch (2011)."""
    # Earth reference values
    ref = 'R' 1.0 'rho' 5.51 'vesc' 11.18 'T' 288.0
    weights = 'R' 0.57 'rho' 1.07 'vesc' 0.70 'T' 2.26

    # Extract planet values
    R = row['pl_rade']
    rho = row['pl_dens'] if pd.notna(row['pl_dens']) else (row['pl_bmasse'] / (R**3)) * 5.51
    vesc = np.sqrt(row['pl_bmasse'] / R) * 11.18 if pd.notna(row['pl_bmasse']) else np.nan
    T = row['pl_eqt']

    if any(pd.isna(x) for x in [R, rho, vesc, T]):
        return np.nan

    # Calculate geometric distance components
```

```

esi_R = 1 - abs((R - ref) / (R + ref))) ** weights
esi_rho = 1 - abs((rho - ref['rho']) / (rho + ref['rho']))) ** weights['rho']
esi_vesc = 1 - abs((vesc - ref['vesc']) / (vesc + ref['vesc']))) ** weights['vesc']
esi_T = 1 - abs((T - ref) / (T + ref))) ** weights

total_weight = sum(weights.values())
esi_global = (esi_R * esi_rho * esi_vesc * esi_T) ** 1 / total_weight

return esi_global

```

7.4 Clustering Core (cluster_pipeline.py)

This module executes the PCA and K-Means operations.

```

Python
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans, AgglomerativeClustering

def run_clustering_pipeline(df, features):
    """Executes PCA and dual clustering to isolate the Terrestrial cluster."""
    # 1. Log transformations for skewed data
    skewed_features = ['pl_bmasse', 'pl_rade', 'pl_insol', 'pl_orbper']
    for feat in skewed_features:
        if feat in features:
            df[f'log_{feat}'] = np.log1p(df[feat])

    # Swap out original features for log features in the feature list
    cluster_features = [f'log_{x}' if x in skewed_features else x for x in features]

    # 2. Extract matrix and handle remaining NaNs (KNN Imputation assumed prior)
    X = df[cluster_features].values

    # 3. Standard Scaling
    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X)

    # 4. PCA Orthogonalization (Targeting 85% variance)

```



```

pca = PCA(n_components=0.85)
X_pca = pca.fit_transform(X_scaled)
print(f"PCA reduced space to {X_pca.shape} components.")

# 5. K-Means Clustering (k=5 based on elbow method)
kmeans = KMeans(n_clusters=5, init='k-means++', n_init=20, random_state=42)
df['KMeans_Cluster'] = kmeans.fit_predict(X_pca)

# 6. Agglomerative Clustering for Cross-Validation on SAME PCA SPACE
ward = AgglomerativeClustering(n_clusters=5, linkage='ward')
df = ward.fit_predict(X_pca)

return df, scaler, pca, kmeans

```

8. Analytical Insights and Independent Validation

A KDD pipeline is only as valuable as its ability to generate profound, actionable insights without relying on the biases of its creators. By subjecting the exoplanetary dataset to strict mathematical constraints and unsupervised clustering, several critical insights emerge regarding both astrophysics and data science methodologies.

8.1 Validation Without Evaluation Leakage

The ultimate test of this pipeline is its ability to recover known, curated candidates for habitability—such as those listed in the Planetary Habitability Laboratory's (PHL) Conservative Catalog⁵⁴—without ever explicitly feeding the algorithm those targets or protecting them from statistical filtration.

When the pipeline is executed, planets such as TRAPPIST-1 e, LHS 1140 b, and TOI-715 b naturally cluster into the identified Terrestrial group based purely on their PCA-reduced structural and thermal vectors. Furthermore, when the ESI and Kopparapu filters are applied post-clustering, these planets consistently rank at the top of the prioritization queue. Because the pipeline guarantees *nothing* by construction, the recovery of these planets serves as genuine, independent validation of the model's predictive power. Conversely, if a well-known "candidate" from early literature fails the clustering or HZ filtering stage, it provides astrophysical insight rather than an algorithmic failure. It suggests that modern archival data has updated the planet's mass or radius, shifting it out of the terrestrial regime and into the volatile-rich sub-Neptune population, thereby contesting its habitability.⁵⁵

8.2 The Venus Conundrum: Distinguishing Structure from Environment

A recurring critique of purely structural clustering is the "Venus Conundrum." Venus and Earth share nearly identical bulk parameters: mass ($0.815M_{\oplus}$ vs $1.0M_{\oplus}$), radius ($0.949R_{\oplus}$ vs

$1.0R_{\oplus}$), and bulk density ($\sim 5.24 \text{ g/cm}^3$ vs 5.51 g/cm^3). If an unsupervised K-Means algorithm is tasked with clustering planets based on these metrics, a Venus-analog and an Earth-analog will be mathematically indistinguishable; they will inevitably occupy the same "Earth-like" structural cluster.

This highlights a profound astrophysical reality: bulk properties do not dictate surface habitability. Habitability is an emergent property dictated by atmospheric photochemistry, geological carbon cycling, and initial volatile inventories.⁵

The revised pipeline solves this conundrum not by forcing the clustering algorithm to artificially separate them, but by layering the deterministic Kopparapu boundaries over the structural cluster. While the K-Means algorithm correctly identifies the Venus-analog as a "Terrestrial Rock," the subsequent HZ filter evaluates its insolation flux. Recognizing that the planet receives flux exceeding the *Recent Venus* limit (and likely the *Runaway Greenhouse* limit), the pipeline decisively deprioritizes the Venus-analog.²⁴ This multi-tiered approach—structural classification followed by thermodynamic filtering—mirrors the actual triage process used by astrobiologists when selecting targets for JWST.

8.3 Salvaging the Radial Velocity Demographic

Perhaps the most significant consequence of implementing the Chen & Kipping (2017) piecewise mass-radius relationship is the dramatic expansion of the searchable parameter space. Traditional data science pipelines that drop rows with missing radii implicitly assume that transit photometry is the only valid discovery method. This systematically erases planets discovered via Radial Velocity (RV) and Microlensing.

Because RV is heavily biased toward detecting massive planets or planets orbiting very close to low-mass stars, and transit photometry is biased toward tightly packed inner systems, the intersection of these two discovery methods leaves a massive gap in the data. By leveraging probabilistic, physics-based imputation, the pipeline reconstructs the missing structural dimensions of RV-discovered super-Earths.¹⁵ This allows the K-Means algorithm to assess a much broader, more demographically complete catalog, uncovering structural similarities between planetary populations that would otherwise remain hidden in fragmented datasets.

9. Conclusion

The transition of exoplanetary science from an era of discovery to an era of detailed atmospheric characterization demands prioritization engines that are both computationally scalable and rigorously bound by physical laws. Prior attempts to apply KDD pipelines to this domain have frequently been compromised by data science anti-patterns: circular evaluation logic, the conflation of collinear variables, arbitrary scoring functions, and the naive deletion of critical Radial Velocity data.

The methodology presented in this report systematically resolves these foundational errors. By retrieving composite planetary parameters programmatically via the NASA Exoplanet Archive TAP service, the pipeline ensures data consistency. The application of the Chen & Kipping (2017) piecewise mass-radius relationship for imputation rescues thousands of demographic data points, maintaining the integrity of the broader exoplanet catalog. Collinearity is elegantly

handled by subjecting the dataset to Principal Component Analysis, ensuring that K-Means and Agglomerative Ward clustering operate on a mathematically orthogonal feature space. Most importantly, the pipeline abandons arbitrary, unweighted scoring mechanisms. By utilizing the standardized Earth Similarity Index (ESI) and the empirically validated Kopparapu Habitable Zone limits as post-clustering filters, the framework cleanly separates planets that are merely structurally similar to Earth from those that actually possess the thermodynamic potential for surface liquid water. By completely eliminating algorithmic "target protection," this pipeline offers genuine, independent predictive power, providing the astronomical community with a robust, reproducible tool for prioritizing the next generation of targets for the James Webb Space Telescope and beyond.

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